

Optimizing Real-time fMRI Neurofeedback to Improve Sustained Attention in ADHD

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Project Aim/Hypothesis

Attention Deficit Hyperactivity Disorder (ADHD) is one of the most studied neurodevelopmental disorders, however the exact nature of this condition remains unclear. There is tremendous heterogeneity across patients with ADHD, which affects 7% of children and often persists into adulthood where ADHD is associated with poor outcomes both socially and academically. Given the high level of societal and personal burden of ADHD and the high rate of side-effects from stimulant medications, identifying new therapy modalities is of potential significant impact. Neurotherapeutics, including neurofeedback and non-invasive brain stimulation have been trialed in ADHD, but in small numbers and with inconsistent findings. Furthermore, the outcomes in recent studies have often focused on executive function instead of clinical symptoms. Recently, significant concern has been raised that much of this heterogeneity in outcomes may come from unoptimized protocols. Nonetheless, fMRI-based neurofeedback techniques have shown significant promise in several neuropsychiatric conditions, including ADHD, and has led to great interest both clinically and across multiple NIH institutes, e.g., the NIMH is holding a Workshop of Neurofeedback Intervention Development on 5/2/2024, and multiple RFAs have recently been posted to foster this growing area of translational research.

The overarching goal of the proposed study is to leverage the natural plasticity of the brain to create new targeted treatments for neurodevelopmental symptoms such as sustained attention in ADHD. This pilot study is focused on optimizing a protocol for real-time functional magnetic resonance imaging neurofeedback (rt-fMRI-NF) within the context of improving sustained attention in ADHD and determining the feasibility of performing rt-fMRI-NF in the Boston Children's Hospital Research Imaging Core with a clinical ADHD population. The specific aims of this pilot study are:

- 1) To iteratively assess the benefit of three methodological parameters:
 - a) Whether it is more effective to provide explicit instructions to participants or allow them to establish their own strategies during neurofeedback.
 - b) Is the optimal control condition for measuring effectiveness: neurofeedback from a *non-implicated* region in the same individual or recorded feedback from the *targeted* region in a different individual in the experimental group.
 - c) Whether presenting visual and/or auditory neural feedback continuously or as an average at the end of each feedback block is more effective.
- 2) To determine the number of sessions needed to demonstrate neuroimaging target engagement and the number of participants needed to demonstrate responder vs. non-responder status.

We hypothesize that: 1) after a brief number of sessions, participants will demonstrate changes in neural activity of the anterior cingulate cortex and potentially in behavioral performance, e.g., increased sustained attention on repeated objective testing, and 2) that the degree of neural modulation and behavioral performance will be modulated or enhanced via the parameters noted above. By assessing change over time, we will be able to determine the range of neurofeedback sessions for each participant that are needed to identify responders vs non-responders and to achieve diminishing returns from repeated sessions. *This pilot study will confirm optimal parameters for this rt-fMRI-NF paradigm and allow for power calculations to determine how many participants are necessary for a subsequent efficacy trial (R01).*

Background

One driving principle of neuroscience is plasticity, the brain's ability to modulate neural activity and connections in temporary and permanent ways. This process can be co-opted through neurofeedback, which is the category of methods that present an individual's own neural signals back to them for the purpose of creating both neural and behavioral changes. This can be done in either healthy individuals or in patients with neuropsychological symptoms and has been proposed as a new and potentially long-lasting avenue for treatment¹. Traditionally, neurofeedback has used signals from electroencephalography (EEG), but this has proven to be highly variable in efficacy, likely due in part to a lack of spatial specificity and inability to measure signals deep in the brain. Over the last decade, methodological and computational advances have allowed the use of functional MRI as a source of neurofeedback signals, which provides superior spatial resolution and whole-brain coverage¹⁻³.

Real-time functional magnetic resonance imaging neurofeedback (rt-fMRI-NF) leverages the rapid calculation of neural activity of a particular region of interest (ROI), a specified neural network, or differential activation or synchrony between multiple ROIs or networks. This signal is then presented back to individuals as comprehensible visual stimuli, which then allows participants to utilize a researcher-provided or self-generated mental strategy to modulate their own neural activity⁴. rt-fMRI-NF has already shown promising results in numerous areas, including emotion regulation⁵, anxiety⁶, and decreasing auditory hallucinations in schizophrenia^{7,8}.

One important therapeutic target for rt-fMRI-NF is the improvement of sustained attention in attention deficit hyperactivity disorder (ADHD). To date, two groups have employed rt-fMRI-NF in individuals with ADHD, with conflicting results⁹⁻¹³, and more research is needed to identify which parameters determine whether rt-fMRI-NF may be an effective treatment for those with impaired attention and executive function. For instance, one group has targeted the dorsal anterior cingulate cortex (dACC) while another has focused on the right inferior frontal gyrus (rIFG). Our own research indicates that the dACC may be more likely to serve as an optimal neurofeedback target, but this is an empirical question that requires testing.

While general guidelines exist for constructing rt-fMRI-NF studies, there are several important parameters which impact efficacy that vary across paradigms and targets². These include whether to provide a specific mental strategy for participants to employ, what is an appropriate sham condition to measure the efficacy of rt-fMRI-NF vs practice and placebo effects, and how best to present feedback to the participants. Identifying the optimal parameters to facilitate rt-fMRI-NF remains a critical point of investigation. In recognition of this, several NIH RFAs have specifically focused on supporting the development of non-invasive neuroplasticity/neurofeedback techniques. *The goal of the current proposal is to optimize an ADHD-focused rt-fMRI-NF protocol and generate the feasibility data necessary to apply for R01 funding specifically focused on testing the efficacy of rt-fMRI-NF to modulate the attention and executive networks of the brain (Aim 1: Target Engagement) and subsequently on sustained attention and clinical symptoms of ADHD (Aim 2: Clinical Outcomes).*

Significance

Upon successful completion of the proposed study, we expect our contribution to be an evidence-based set of optimal parameters for rt-fMRI-NF that both engages and successfully increases dACC activity during a sustained attention task. This contribution is expected to be significant because it will demonstrate target engagement and allow the design and execution of an appropriately powered, hypothesis-driven, rt-fMRI-NF efficacy trial for improving sustained attention in participants with ADHD. Long-term, the establishment of rt-fMRI-NF as an effective non-pharmacological treatment option for individuals with ADHD and other neurodevelopmental conditions may lead to the identification of more generalizable treatment techniques that co-opt the same neural plasticity mechanisms without requiring frequent MRI sessions for each patient.

Methods

Study Protocol. This study will include the collection and analysis of prospectively acquired neuroimaging and behavioral data from typically developing participants 12 years of age or older with ADHD who are able to follow verbal and written directions in their native language. Participants will be recruited from the general community, the PI's clinical practice at BCH (about 100 unique ADHD patients seen annually), and from the Translational Neuroscience Center's Research Participant Registry (RPR). Participants will undergo several testing/training and MRI visits. During the first visit, participants will complete a series of computer/paper tasks that are designed to estimate behavioral variables such as sustained attention ability. Participants will also be asked to fill out brief questionnaires to estimate pubertal stage and handedness. Additionally, participants with ADHD who have not recently undergone diagnostic testing with a research reliable evaluator will be asked to complete this testing during the first visit. We will also explain what happens during neurofeedback training and may provide strategies for modulating brain activity and improving behavioral performance before participants enter the scanner. Finally, during the first or second visit, subjects will participate in their first MRI session, which will involve resting-state runs and neurofeedback runs, during which functional data will be acquired while the participant completes a sustained attention task while viewing a visual neurofeedback signal based on their own brain activity in the neural correlate of interest, presented in a developmentally appropriate and comprehensible way. rt-fMRI-NF will be implemented using the RT-Cloud containerized software package that

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